

Beyond Marginal Guarantees: A Summary Report of the Full Book

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Abstract—This document summarizes *Beyond Marginal Guarantees*, a 237-page, twelve-chapter book-length report examining whether conformal prediction's marginal coverage guarantee holds within specific operating regimes of a hospital emergency department, rather than only on average. The book trains a gradient-boosting surrogate on a calibrated discrete-event simulation of ED operations and compares standard split conformal prediction against Mondrian conformal prediction across nine staffing x arrival-rate categories. Its central finding is that standard conformal prediction's marginal guarantee conceals a coverage collapse to 68.2% in the single highest-stakes operating regime - understaffed and high-arrival - which Mondrian conformal prediction corrects to 90.9-92.0%, a result confirmed significant across 30 repeated trials and replicated at an independent second hospital department. Beyond this core result, the book develops five further conformal methods, stress-tests the exchangeability assumption to destruction, benchmarks five surrogate architectures, and translates the findings into an operational staffing dashboard - each derivation, implementation detail, and honest limitation documented in full. This report exists as a navigational guide to that material: what each chapter covers, where the central result is proven, and what a reader can safely skip.

Index Terms—Book summary, conformal prediction, Mondrian conformal prediction, discrete-event simulation, queueing theory, emergency department operations, reproducibility.

I. PURPOSE OF THIS SUMMARY

At the faculty guide's request, this project's original short course report was expanded into a book-length treatment: 237 pages across twelve numbered chapters plus two appendices, formatted as a technical textbook - decimal-numbered headings, chapter-scoped figure/table/equation numbering, and IEEE-style bracketed citations. A condensed, 8-page conference-style paper covering the same central finding also exists as a companion document. This report is neither of those - it is a short guide to the book: what each chapter contains, where its central result is actually proven, and which chapters a reader with limited time can defer or skip.

II. HOW THE BOOK IS ORGANIZED

TABLE I
BOOK STRUCTURE AND CHAPTER STARTING PAGES

Ch.	Title	Page
1	The ER Uncertainty Dilemma	11
2	Foundations and Shadows (Literature Review)	21
3	The Marginal Trap (Research Gap)	47
4	Architecture of Conformal Simulation (Methodology)	55
5	Software Engineering and Reproducibility	89
6	Beyond Standard CP (Five Extension Methods)	98
7	Empirical Validation and Metamodel Benchmarking	111
8	When Exchangeability Fails	136
9	Cross-Site Generalization	144
10	Translational Health Operations	152
11	Synthesis and Uncharted Horizons	161
12	References	166

A. Chapters 1-3: The Problem

Chapter 1 motivates why a point-prediction surrogate needs a calibrated interval rather than a single number, and why Emergency Department operations - discrete,

stochastic, queueing-driven - are a structurally different test domain than the physics simulations conformal prediction was first validated on (Section 1.4, citing Gopakumar et al. [1]). Chapter 2 is the book's literature review in full depth (26 pages, Sections 2.1-2.8): conformal prediction foundations, Mondrian and conditional-coverage methods [2], surrogate modeling and UQ, queueing-theoretic ED operations research, and discrete-event ED simulation, closing with a critical assessment (2.7) of where this project sits relative to prior work. Chapter 3 states the research gap directly and formalizes the operational-risk framing (3.6) that motivates treating conditional, not only marginal, coverage as the quantity that matters.

B. Chapters 4-5: Methodology and Implementation

Chapter 4 derives every theoretical result later chapters rely on: the discrete-event simulation design, offered-load capacity sizing (4.2.1), all five uncertainty-quantification methods with formal algorithm statements (4.4.5), and the heavy-traffic and quantile-monotonicity proofs behind them (4.7). Chapter 5 documents the software itself - dataset, module-by-module code walkthrough, and reproducibility practice - matched by the full source-code listings in Appendix A.

C. Chapter 6: Five Extensions to Standard CP

Conformalized quantile regression and Mondrian-CQR (6.1), conformal risk control for bounding operational overflow severity rather than only its probability (6.2), adaptive conformal inference under a live demand-surge stream (6.3), likelihood-ratio weighted CP under covariate shift (6.4), and a new queueing-theoretic normalized CP - QT-CP - grounded in Kingman's heavy-traffic approximation [3] (6.5).

D. Chapters 7-9: Results

Chapter 7 is the book's central empirical chapter: the coverage collapse and its correction (7.5), full per-category breakdowns for all three affected targets (7.5.2-7.5.3), the mechanism behind why the understaffed/high-demand category specifically fails (7.5.4), and 30-repeat statistical significance testing (7.6). Chapter 8 stress-tests the exchangeability assumption every method here depends on, to the point of collapse. Chapter 9 replicates the entire pipeline - simulation calibration, surrogate training, Mondrian binning - independently at a second hospital department.

E. Chapters 10-11: Application and Synthesis

Chapter 10 translates the statistical result into an operational prediction-interval dashboard and a capacity-planning optimization built on this project's own conformal intervals. Chapter 11 synthesizes the contributions, states open theoretical questions, and lays out a roadmap for future work, including combining QT-CP's continuous scale with Mondrian's discrete binning.

III. THE CENTRAL FINDING

Averaged across the full test distribution, standard and Mondrian CP both meet the 90% marginal coverage target (Chapter 7.4-7.5). Evaluated per category instead of pooled, standard CP's coverage in the single understaffed / high-arrival-rate category collapses to 68.2% - 22 points under target - a failure the marginal average conceals entirely. Mondrian CP restores 90.9-92.0% coverage in that same category (Table II), a difference significant under a paired t-test across 30 independent repeats ($p < 0.001$, Section 7.6) and replicated at an independent second department, where the same category's coverage improves from 76.2% to 89.3% (Chapter 9.3).

TABLE II
WORST-CATEGORY COVERAGE, UNDERSTAFFED / HIGH-ARRIVAL

Site	Pooled CP	Mondrian CP
Dept. A (primary)	68.2%	90.9%
Dept. B (replication)	76.2%	89.3%

IV. WHAT THE BOOK ADDS BEYOND THE COMPANION PAPER

The 8-page companion paper states the same central finding and the same five extension methods in condensed form. The book's additional length is not padding: it carries the full derivations and formal proofs behind every method (Section 4.7), a literature review roughly six times longer than the paper's Related Work section (Chapter 2), complete software documentation with a full source-code appendix (Chapter 5, Appendix A), per-category result tables broken out for every target rather than summarized for space (Appendix B), a dedicated operational chapter building an actual staffing dashboard and capacity optimizer that the paper only mentions as future work (Chapter 10), and an

extended synthesis of open questions and a concrete future-work roadmap (Chapter 11).

V. HONEST RESULTS AND LIMITATIONS

Not every result in the book is a clean win, and it is written to say so directly. QT-CP (6.5) achieves narrower average intervals than standard CP on three of four targets but does not match Mondrian CP's worst-category coverage on any target, and underperforms even standard CP on n_{patients} - the one target with no real conditional gap to correct in the first place (7.5.6). Both standard and Mondrian CP's coverage advantage collapses together under severe exchangeability violation (Chapter 8): Mondrian's per-category structure offers no protection once test points fall entirely outside every calibration category's own support. Service-time distributions are literature-calibrated log-normals rather than recoverable from the source dataset, and cross-site replication covers two departments from one health system - both disclosed explicitly in Chapter 11.4 rather than left for a reader to discover.

VI. HOW TO READ THE BOOK

Readers primarily interested in the empirical findings rather than the underlying theory may proceed directly to Chapter 7 after Chapter 3 - the book's own Preface gives this same guidance. Readers verifying the project's engineering rather than its statistics should read Chapter 5 and Appendix A. Readers evaluating the report for a specific operational deployment should read Chapters 9 and 10 for generalizability and translation to practice, and Chapter 11.4 for what remains unresolved.

VII. CONCLUSION

Beyond Marginal Guarantees documents, at book length, a single central claim also stated in the companion paper: a marginal coverage guarantee can conceal a severe conditional failure in exactly the operating regime a decision-maker cares most about, and Mondrian conformal prediction corrects it without sacrificing the marginal guarantee. What the extra 200-odd pages contribute is depth - full proofs, full per-category evidence, full reproducibility - and honesty about where the project's own extensions (QT-CP) and assumptions (exchangeability) do not fully hold up.

REFERENCES

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- [3] J. F. C. Kingman, "On Queues in Heavy Traffic," *J. R. Stat. Soc. Ser. B*, vol. 24, no. 2, pp. 383-392, 1962.